Uncertainty management in rule-based expert systems

3

In which we present the main uncertainty management paradigms, Bayesian reasoning and certainty factors, discuss their relative merits and consider examples to illustrate the theory.

3.1 Introduction, or what is uncertainty?

One of the common characteristics of the information available to human experts is its imperfection. Information can be incomplete, inconsistent, uncertain, or all three. In other words, information is often unsuitable for solving a problem. However, an expert can cope with these defects and can usually make correct judgements and right decisions. Expert systems also have to be able to handle uncertainty and draw valid conclusions.

What is uncertainty in expert systems?

Uncertainty can be defined as the lack of the exact knowledge that would enable us to reach a perfectly reliable conclusion (Stephanou and Sage, 1987). Classical logic permits only exact reasoning. It assumes that perfect knowledge always exists and the law of the excluded middle can always be applied:

IF *A* is true
THEN *A* is not false

and

IF *B* is false
THEN *B* is not true

Unfortunately most real-world problems where expert systems could be used do not provide us with such clear-cut knowledge. The available information often contains inexact, incomplete or even unmeasurable data.

What are the sources of uncertain knowledge in expert systems?

In general, we can identify four main sources: weak implications, imprecise language, unknown data, and the difficulty of combining the views of different

experts (Bonissone and Tong, 1985). Let us consider these sources in more detail.

- **Weak implications**. Rule-based expert systems often suffer from weak implications and vague associations. Domain experts and knowledge engineers have the painful, and rather hopeless, task of establishing concrete correlations between IF (condition) and THEN (action) parts of the rules. Therefore, expert systems need to have the ability to handle vague associations, for example by accepting the degree of correlations as numerical certainty factors.
- **Imprecise language**. Our natural language is inherently ambiguous and imprecise. We describe facts with such terms as **often** and **sometimes**, **frequently** and **hardly ever**. As a result, it can be difficult to express knowledge in the precise IF-THEN form of production rules. However, if the meaning of the facts is quantified, it can be used in expert systems. In 1944, Ray Simpson asked 355 high school and college students to place 20 terms like **often** on a scale between 1 and 100 (Simpson, 1944). In 1968, Milton Hakel repeated this experiment (Hakel, 1968). Their results are presented in Table 3.1.

  Quantifying the meaning of the terms enables an expert system to establish an appropriate matching of the IF (condition) part of the rules with facts available in the database.
- **Unknown data**. When the data is incomplete or missing, the only solution is to accept the value 'unknown' and proceed to an approximate reasoning with this value.
- **Combining the views of different experts**. Large expert systems usually combine the knowledge and expertise of a number of experts. For example, nine experts participated in the development of PROSPECTOR, an expert system for mineral exploration (Duda *et al.*, 1979). Unfortunately, experts seldom reach exactly the same conclusions. Usually, experts have contradictory opinions and produce conflicting rules. To resolve the conflict, the knowledge engineer has to attach a weight to each expert and then calculate the composite conclusion. However, even a domain expert generally does not have the same uniform level of expertise throughout a domain. In addition, no systematic method exists to obtain weights.

In summary, an expert system should be able to manage uncertainties because any real-world domain contains inexact knowledge and needs to cope with incomplete, inconsistent or even missing data. A number of numeric and non-numeric methods have been developed to deal with uncertainty in rule-based expert systems (Bhatnagar and Kanal, 1986). In this chapter, we consider the most popular uncertainty management paradigms: Bayesian reasoning and certainty factors. However, we first look at the basic principles of classical probability theory.

Table 3.1 Quantification of ambiguous and imprecise terms on a time–frequency scale

Ray Simpson (1944)		Milton Hakel (1968)	
Term	**Mean value**	**Term**	**Mean value**
Always	99	Always	100
Very often	88	Very often	87
Usually	85	Usually	79
Often	78	Often	74
Generally	78	Rather often	74
Frequently	73	Frequently	72
Rather often	65	Generally	72
About as often as not	50	About as often as not	50
Now and then	20	Now and then	34
Sometimes	20	Sometimes	29
Occasionally	20	Occasionally	28
Once in a while	15	Once in a while	22
Not often	13	Not often	16
Usually not	10	Usually not	16
Seldom	10	Seldom	9
Hardly ever	7	Hardly ever	8
Very seldom	6	Very seldom	7
Rarely	5	Rarely	5
Almost never	3	Almost never	2
Never	0	Never	0

3.2 Basic probability theory

The basic concept of probability plays a significant role in our everyday life. We try to determine the probability of rain and the prospects of our promotion, the odds that the Australian cricket team will win the next test match, and the likelihood of winning a million dollars in Tattslotto.

The concept of probability has a long history that goes back thousands of years when words like 'probably', 'likely', 'maybe', 'perhaps' and 'possibly' were introduced into spoken languages (Good, 1959). However, the mathematical theory of probability was formulated only in the 17th century.

How can we define probability?

The probability of an event is the proportion of cases in which the event occurs (Good, 1959). Probability can also be defined as a scientific measure of chance. Detailed analysis of modern probability theory can be found in such well-known textbooks as Feller (1966; 1968) and Fine (1973). In this chapter, we examine only the basic ideas used in representing uncertainties in expert systems.

Probability can be expressed mathematically as a numerical index with a range between zero (an absolute impossibility) to unity (an absolute certainty). Most events have a probability index strictly between 0 and 1, which means that

each event has **at least** two possible outcomes: favourable outcome or success, and unfavourable outcome or failure.

The probability of success and failure can be determined as follows:

$$P(\text{success}) = \frac{\text{the number of successes}}{\text{the number of possible outcomes}} \tag{3.1}$$

$$P(\text{failure}) = \frac{\text{the number of failures}}{\text{the number of possible outcomes}} \tag{3.2}$$

Therefore, if s is the number of times success can occur, and f is the number of times failure can occur, then

$$P(\text{success}) = p = \frac{s}{s+f} \tag{3.3}$$

$$P(\text{failure}) = q = \frac{f}{s+f} \tag{3.4}$$

and

$$p + q = 1 \tag{3.5}$$

Let us consider classical examples with a coin and a dice. If we throw a coin, the probability of getting a head will be equal to the probability of getting a tail. In a single throw, $s = f = 1$, and therefore the probability of getting a head (or a tail) is 0.5.

Consider now a dice and determine the probability of getting a 6 from a single throw. If we assume a 6 as the only success, then $s = 1$ and $f = 5$, since there is just one way of getting a 6, and there are five ways of not getting a 6 in a single throw. Therefore, the probability of getting a 6 is

$$p = \frac{1}{1+5} = 0.1666$$

and the probability of not getting a 6 is

$$q = \frac{5}{1+5} = 0.8333$$

So far, we have been concerned with events that are independent and mutually exclusive (i.e. events that cannot happen simultaneously). In the dice experiment, the two events of obtaining a 6 and, for example, a 1 are mutually exclusive because we cannot obtain a 6 **and** a 1 simultaneously in a single throw. However, events that are not independent may affect the likelihood of one or the other occurring. Consider, for instance, the probability of getting a 6 in a

single throw, knowing this time that a 1 has not come up. There are still five ways of not getting a 6, but one of them can be eliminated as we know that a 1 has not been obtained. Thus,

$$p = \frac{1}{1 + (5 - 1)}$$

Let A be an event in the world and B be another event. Suppose that events A and B are not mutually exclusive, but occur conditionally on the occurrence of the other. The probability that event A will occur if event B occurs is called the **conditional probability**. Conditional probability is denoted mathematically as $p(A|B)$ in which the vertical bar represents GIVEN and the complete probability expression is interpreted as 'Conditional probability of event A occurring given that event B has occurred'.

$$p(A|B) = \frac{\text{the number of times } A \text{ and } B \text{ can occur}}{\text{the number of times } B \text{ can occur}} \tag{3.6}$$

The number of times A and B can occur, or the probability that both A and B will occur, is called the **joint probability** of A and B. It is represented mathematically as $p(A \cap B)$. The number of ways B can occur is the probability of B, $p(B)$, and thus

$$p(A|B) = \frac{p(A \cap B)}{p(B)} \tag{3.7}$$

Similarly, the conditional probability of event B occurring given that event A has occurred equals

$$p(B|A) = \frac{p(B \cap A)}{p(A)} \tag{3.8}$$

Hence,

$$p(B \cap A) = p(B|A) \times p(A) \tag{3.9}$$

The joint probability is commutative, thus

$$p(A \cap B) = p(B \cap A)$$

Therefore,

$$p(A \cap B) = p(B|A) \times p(A) \tag{3.10}$$

Substituting Eq. (3.10) into Eq. (3.7) yields the following equation:

$$p(A|B) = \frac{p(B|A) \times p(A)}{p(B)} \tag{3.11}$$

where:

$p(A|B)$ is the conditional probability that event A occurs given that event B has occurred;

$p(B|A)$ is the conditional probability of event B occurring given that event A has occurred;

$p(A)$ is the probability of event A occurring;

$p(B)$ is the probability of event B occurring.

Equation (3.11) is known as the **Bayesian rule**, which is named after Thomas Bayes, an 18th-century British mathematician who introduced this rule.

The concept of conditional probability introduced so far considered that event A was dependent upon event B. This principle can be extended to event A being dependent on a number of mutually exclusive events $B_1, B_2, \ldots, B_n$. The following set of equations can then be derived from Eq. (3.7):

$$\begin{aligned} p(A \cap B_1) &= p(A|B_1) \times p(B_1) \\ p(A \cap B_2) &= p(A|B_2) \times p(B_2) \\ &\vdots \\ p(A \cap B_n) &= p(A|B_n) \times p(B_n) \end{aligned}$$

or when combined:

$$\sum_{i=1}^{n} p(A \cap B_i) = \sum_{i=1}^{n} p(A|B_i) \times p(B_i) \tag{3.12}$$

If Eq. (3.12) is summed over an exhaustive list of events for B_i as illustrated in Figure 3.1, we obtain

$$\sum_{i=1}^{n} p(A \cap B_i) = p(A) \tag{3.13}$$

It reduces Eq. (3.12) to the following conditional probability equation:

$$p(A) = \sum_{i=1}^{n} p(A|B_i) \times p(B_i) \tag{3.14}$$

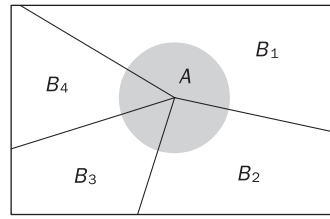


Figure 3.1 The joint probability

If the occurrence of event A depends on only two mutually exclusive events, that is B and NOT B, then Eq. (3.14) becomes

$$p(A) = p(A|B) \times p(B) + p(A|\neg B) \times p(\neg B) \tag{3.15}$$

where $\neg$ is the logical function NOT.

Similarly,

$$p(B) = p(B|A) \times p(A) + p(B|\neg A) \times p(\neg A) \tag{3.16}$$

Let us now substitute Eq. (3.16) into the Bayesian rule (3.11) to yield

$$p(A|B) = \frac{p(B|A) \times p(A)}{p(B|A) \times p(A) + p(B|\neg A) \times p(\neg A)} \tag{3.17}$$

Equation (3.17) provides the background for the application of probability theory to manage uncertainty in expert systems.

3.3 Bayesian reasoning

With Eq. (3.17) we can now leave basic probability theory and turn our attention back to expert systems. Suppose all rules in the knowledge base are represented in the following form:

IF *E* is true
THEN *H* is true {with probability *p*}

This rule implies that if event E occurs, then the probability that event H will occur is p.

What if event *E* has occurred but we do not know whether event *H* has occurred? Can we compute the probability that event *H* has occurred as well?

Equation (3.17) tells us how to do this. We simply use H and E instead of A and B. In expert systems, H usually represents a hypothesis and E denotes evidence to

support this hypothesis. Thus, Eq. (3.17) expressed in terms of hypotheses and evidence looks like this (Firebaugh, 1989):

$$p(H|E) = \frac{p(E|H) \times p(H)}{p(E|H) \times p(H) + p(E|\neg H) \times p(\neg H)} \tag{3.18}$$

where:

$p(H)$ is the prior probability of hypothesis H being true;

$p(E|H)$ is the probability that hypothesis H being true will result in evidence E;

$p(\neg H)$ is the prior probability of hypothesis H being false;

$p(E|\neg H)$ is the probability of finding evidence E even when hypothesis H is false.

Equation (3.18) suggests that the probability of hypothesis H, $p(H)$, has to be defined before any evidence is examined. In expert systems, the probabilities required to solve a problem are provided by experts. An expert determines the prior probabilities for possible hypotheses $p(H)$ and $p(\neg H)$, and also the conditional probabilities for observing evidence E if hypothesis H is true, $p(E|H)$, and if hypothesis H is false, $p(E|\neg H)$. Users provide information about the evidence observed and the expert system computes $p(H|E)$ for hypothesis H in light of the user-supplied evidence E. Probability $p(H|E)$ is called the **posterior probability** of hypothesis H upon observing evidence E.

What if the expert, based on single evidence E, cannot choose a single hypothesis but rather provides multiple hypotheses $H_1, H_2, \ldots, H_m$? Or given multiple evidences $E_1, E_2, \ldots, E_n$, the expert also produces multiple hypotheses?

We can generalise Eq. (3.18) to take into account both multiple hypotheses $H_1, H_2, \ldots, H_m$ and multiple evidences $E_1, E_2, \ldots, E_n$. But the hypotheses as well as the evidences must be mutually exclusive and exhaustive.

Single evidence E and multiple hypotheses $H_1, H_2, \ldots, H_m$ follow:

$$p(H_i|E) = \frac{p(E|H_i) \times p(H_i)}{\sum_{k=1}^{m} p(E|H_k) \times p(H_k)} \tag{3.19}$$

Multiple evidences $E_1, E_2, \ldots, E_n$ and multiple hypotheses $H_1, H_2, \ldots, H_m$ follow:

$$p(H_i|E_1E_2 \ldots E_n) = \frac{p(E_1E_2 \ldots E_n|H_i) \times p(H_i)}{\sum_{k=1}^{m} p(E_1E_2 \ldots E_n|H_k) \times p(H_k)} \tag{3.20}$$

An application of Eq. (3.20) requires us to obtain the conditional probabilities of all possible combinations of evidences for all hypotheses. This requirement

places an enormous burden on the expert and makes his or her task practically impossible. Therefore, in expert systems, subtleties of evidence should be suppressed and conditional independence among different evidences assumed (Ng and Abramson, 1990). Thus, instead of unworkable Eq. (3.20), we attain

$$p(H_i|E_1E_2\ldots E_n) = \frac{p(E_1|H_i) \times p(E_2|H_i) \times \ldots \times p(E_n|H_i) \times p(H_i)}{\sum_{k=1}^{m} p(E_1|H_k) \times p(E_2|H_k) \times \ldots \times p(E_n|H_k) \times p(H_k)} \tag{3.21}$$

How does an expert system compute all posterior probabilities and finally rank potentially true hypotheses?

Let us consider a simple example. Suppose an expert, given three conditionally independent evidences E_1, E_2 and E_3, creates three mutually exclusive and exhaustive hypotheses H_1, H_2 and H_3, and provides prior probabilities for these hypotheses – $p(H_1)$, $p(H_2)$ and $p(H_3)$, respectively. The expert also determines the conditional probabilities of observing each evidence for all possible hypotheses. Table 3.2 illustrates the prior and conditional probabilities provided by the expert.

Assume that we first observe evidence E_3. The expert system computes the posterior probabilities for all hypotheses according to Eq. (3.19):

$$p(H_i|E_3) = \frac{p(E_3|H_i) \times p(H_i)}{\sum_{k=1}^{3} p(E_3|H_k) \times p(H_k)}, \qquad i = 1, 2, 3$$

Thus,

$$p(H_1|E_3) = \frac{0.6 \times 0.40}{0.6 \times 0.40 + 0.7 \times 0.35 + 0.9 \times 0.25} = 0.34$$

$$p(H_2|E_3) = \frac{0.7 \times 0.35}{0.6 \times 0.40 + 0.7 \times 0.35 + 0.9 \times 0.25} = 0.34$$

$$p(H_3|E_3) = \frac{0.9 \times 0.25}{0.6 \times 0.40 + 0.7 \times 0.35 + 0.9 \times 0.25} = 0.32$$

Table 3.2 The prior and conditional probabilities

	Hypothesis		
Probability	$i = 1$	$i = 2$	$i = 3$
$p(H_i)$	0.40	0.35	0.25
$p(E_1\|H_i)$	0.3	0.8	0.5
$p(E_2\|H_i)$	0.9	0.0	0.7
$p(E_3\|H_i)$	0.6	0.7	0.9

As you can see, after evidence E_3 is observed, belief in hypothesis H_1 decreases and becomes equal to belief in hypothesis H_2. Belief in hypothesis H_3 increases and even nearly reaches beliefs in hypotheses H_1 and H_2.

Suppose now that we observe evidence E_1. The posterior probabilities are calculated by Eq. (3.21):

$$p(H_i|E_1E_3) = \frac{p(E_1|H_i) \times p(E_3|H_i) \times p(H_i)}{\sum_{k=1}^{3} p(E_1|H_k) \times p(E_3|H_k) \times p(H_k)}, \qquad i = 1, 2, 3$$

Hence,

$$p(H_1|E_1E_3) = \frac{0.3 \times 0.6 \times 0.40}{0.3 \times 0.6 \times 0.40 + 0.8 \times 0.7 \times 0.35 + 0.5 \times 0.9 \times 0.25} = 0.19$$

$$p(H_2|E_1E_3) = \frac{0.8 \times 0.7 \times 0.35}{0.3 \times 0.6 \times 0.40 + 0.8 \times 0.7 \times 0.35 + 0.5 \times 0.9 \times 0.25} = 0.52$$

$$p(H_3|E_1E_3) = \frac{0.5 \times 0.9 \times 0.25}{0.3 \times 0.6 \times 0.40 + 0.8 \times 0.7 \times 0.35 + 0.5 \times 0.9 \times 0.25} = 0.29$$

Hypothesis H_2 is now considered as the most likely one, while belief in hypothesis H_1 has decreased dramatically.

After observing evidence E_2 as well, the expert system calculates the final posterior probabilities for all hypotheses:

$$p(H_i|E_1E_2E_3) = \frac{p(E_1|H_i) \times p(E_2|H_i) \times p(E_3|H_i) \times p(H_i)}{\sum_{k=1}^{3} p(E_1|H_k) \times p(E_2|H_k) \times p(E_3|H_k) \times p(H_k)}, \qquad i = 1, 2, 3$$

Thus,

$$p(H_1|E_1E_2E_3) = \frac{0.3 \times 0.9 \times 0.6 \times 0.40}{0.3 \times 0.9 \times 0.6 \times 0.40 + 0.8 \times 0.0 \times 0.7 \times 0.35 + 0.5 \times 0.7 \times 0.9 \times 0.25} = 0.45$$

$$p(H_2|E_1E_2E_3) = \frac{0.8 \times 0.0 \times 0.7 \times 0.35}{0.3 \times 0.9 \times 0.6 \times 0.40 + 0.8 \times 0.0 \times 0.7 \times 0.35 + 0.5 \times 0.7 \times 0.9 \times 0.25} = 0$$

$$p(H_3|E_1E_2E_3) = \frac{0.5 \times 0.7 \times 0.9 \times 0.25}{0.3 \times 0.9 \times 0.6 \times 0.40 + 0.8 \times 0.0 \times 0.7 \times 0.35 + 0.5 \times 0.7 \times 0.9 \times 0.25} = 0.55$$

Although the initial ranking provided by the expert was H_1, H_2 and H_3, only hypotheses H_1 and H_3 remain under consideration after all evidences

(E_1, E_2 and E_3) were observed. Hypothesis H_2 can now be completely abandoned. Note that hypothesis H_3 is considered more likely than hypothesis H_1.

PROSPECTOR, an expert system for mineral exploration, was the first system to use Bayesian rules of evidence to compute $p(H|E)$ and propagate uncertainties throughout the system (Duda *et al.*, 1979). To help interpret Bayesian reasoning in expert systems, consider a simple example.

3.4 FORECAST: Bayesian accumulation of evidence

Let us develop an expert system for a real problem such as the weather forecast. Our expert system will be required to work out if it is going to rain tomorrow. It will need some real data, which can be obtained from the weather bureau.

Table 3.3 summarises London weather for March 1982. It gives the minimum and maximum temperatures, rainfall and sunshine for each day. If rainfall is zero it is a dry day.

The expert system should give us two possible outcomes – *tomorrow is rain* and *tomorrow is dry* – and provide their likelihood. In other words, the expert system must determine the conditional probabilities of the two hypotheses *tomorrow is rain* and *tomorrow is dry*.

To apply the Bayesian rule (3.18), we should provide the prior probabilities of these hypotheses.

The first thing to do is to write two basic rules that, with the data provided, could predict the weather for tomorrow.

```
Rule: 1
IF      today is rain
THEN    tomorrow is rain

Rule: 2
IF      today is dry
THEN    tomorrow is dry
```

Using these rules we will make only 10 mistakes – every time a wet day precedes a dry one, or a dry day precedes a wet one. Thus, we can accept the prior probabilities of 0.5 for both hypotheses and rewrite our rules in the following form:

```
Rule: 1
IF      today is rain {LS 2.5 LN .6}
THEN    tomorrow is rain {prior .5}

Rule: 2
IF      today is dry {LS 1.6 LN .4}
THEN    tomorrow is dry {prior .5}
```

Table 3.3 London weather summary for March 1982

Day of month	Min. temp. °C	Max. temp. °C	Rainfall mm	Sunshine hours	Actual weather	Weather forecast
1	9.4	11.0	17.5	3.2	Rain	–
2	4.2	12.5	4.1	6.2	Rain	Rain
3	7.6	11.2	7.7	1.1	Rain	Rain
4	5.7	10.5	0.0	4.3	Dry	Rain*
5	3.0	12.0	0.0	9.5	Dry	Dry
6	4.4	9.6	0.0	3.5	Dry	Dry
7	4.8	9.4	4.6	10.1	Rain	Rain
8	1.8	9.2	5.5	7.8	Rain	Rain
9	2.4	10.2	4.8	4.1	Rain	Rain
10	5.5	12.7	4.2	3.8	Rain	Rain
11	3.7	10.9	4.4	9.2	Rain	Rain
12	5.9	10.0	4.8	7.1	Rain	Rain
13	3.0	11.9	0.0	8.3	Dry	Rain*
14	5.4	12.1	4.8	1.8	Rain	Dry*
15	8.8	9.1	8.8	0.0	Rain	Rain
16	2.4	8.4	3.0	3.1	Rain	Rain
17	4.3	10.8	0.0	4.3	Dry	Dry
18	3.4	11.1	4.2	6.6	Rain	Rain
19	4.4	8.4	5.4	0.7	Rain	Rain
20	5.1	7.9	3.0	0.1	Rain	Rain
21	4.4	7.3	0.0	0.0	Dry	Dry
22	5.6	14.0	0.0	6.8	Dry	Dry
23	5.7	14.0	0.0	8.8	Dry	Dry
24	2.9	13.9	0.0	9.5	Dry	Dry
25	5.8	16.4	0.0	10.3	Dry	Dry
26	3.9	17.0	0.0	9.9	Dry	Dry
27	3.8	18.3	0.0	8.3	Dry	Dry
28	5.8	15.4	3.2	7.0	Rain	Dry*
29	6.7	8.8	0.0	4.2	Dry	Dry
30	4.5	9.6	4.8	8.8	Rain	Rain
31	4.6	9.6	3.2	4.2	Rain	Rain

* Errors in weather forecast.

The value of LS represents a measure of the expert belief in hypothesis H if evidence E is present. It is called **likelihood of sufficiency**. The likelihood of sufficiency is defined as the ratio of $p(E|H)$ over $p(E|\neg H)$

$$LS = \frac{p(E|H)}{p(E|\neg H)} \tag{3.22}$$

In our case, LS is the probability of getting rain today if we have rain tomorrow, divided by the probability of getting rain today if there is no rain tomorrow:

$$LS = \frac{p(today\ is\ rain\,|\,tomorrow\ is\ rain)}{p(today\ is\ rain\,|\,tomorrow\ is\ dry)}$$

LN, as you may already have guessed, is a measure of discredit to hypothesis *H* if evidence *E* is missing. *LN* is called **likelihood of necessity** and defined as

$$LN = \frac{p(\neg E|H)}{p(\neg E|\neg H)} \tag{3.23}$$

In our weather example, *LN* is the probability of not getting rain today if we have rain tomorrow, divided by the probability of not getting rain today if there is no rain tomorrow:

$$LN = \frac{p(today\ is\ dry\,|\,tomorrow\ is\ rain)}{p(today\ is\ dry\,|\,tomorrow\ is\ dry)}$$

Note that *LN* cannot be derived from *LS*. The domain expert must provide both values independently.

How does the domain expert determine values of the likelihood of sufficiency and the likelihood of necessity? Is the expert required to deal with conditional probabilities?

To provide values for *LS* and *LN*, an expert does not need to determine exact values of conditional probabilities. The expert decides likelihood ratios directly. High values of *LS* ($LS \gg 1$) indicate that the rule strongly supports the hypothesis if the evidence is observed, and low values of *LN* ($0 < LN < 1$) suggest that the rule also strongly opposes the hypothesis if the evidence is missing.

Since the conditional probabilities can be easily computed from the likelihood ratios *LS* and *LN*, this approach can use the Bayesian rule to propagate evidence.

Go back now to the London weather. Rule 1 tells us that if it is raining today, there is a high probability of rain tomorrow ($LS = 2.5$). But even if there is no rain today, or in other words today is dry, there is still some chance of having rain tomorrow ($LN = 0.6$).

Rule 2, on the other hand, clarifies the situation with a dry day. If it is dry today, then the probability of a dry day tomorrow is also high ($LS = 1.6$). However, as you can see, the probability of rain tomorrow if it is raining today is higher than the probability of a dry day tomorrow if it is dry today. Why? The values of *LS* and *LN* are usually determined by the domain expert. In our weather example, these values can also be confirmed from the statistical information published by the weather bureau. Rule 2 also determines the chance of a dry day tomorrow even if today we have rain ($LN = 0.4$).

How does the expert system get the overall probability of a dry or wet day tomorrow?

In the rule-based expert system, the prior probability of the consequent, $p(H)$, is converted into the prior odds:

$$O(H) = \frac{p(H)}{1 - p(H)} \tag{3.24}$$

The prior probability is only used when the uncertainty of the consequent is adjusted for the first time. Then in order to obtain the posterior odds, the prior odds are updated by *LS* if the antecedent of the rule (in other words, evidence) is true and by *LN* if the antecedent is false:

$$O(H|E) = LS \times O(H) \tag{3.25}$$

and

$$O(H|\neg E) = LN \times O(H) \tag{3.26}$$

The posterior odds are then used to recover the posterior probabilities:

$$p(H|E) = \frac{O(H|E)}{1 + O(H|E)} \tag{3.27}$$

and

$$p(H|\neg E) = \frac{O(H|\neg E)}{1 + O(H|\neg E)} \tag{3.28}$$

Our London weather example shows how this scheme works. Suppose the user indicates that *today is rain*. Rule 1 is fired and the prior probability of *tomorrow is rain* is converted into the prior odds:

$$O(\textit{tomorrow is rain}) = \frac{0.5}{1 - 0.5} = 1.0$$

The evidence *today is rain* increases the odds by a factor of 2.5, thereby raising the probability from 0.5 to 0.71:

$$O(\textit{tomorrow is rain} \,|\, \textit{today is rain}) = 2.5 \times 1.0 = 2.5$$

$$p(\textit{tomorrow is rain} \,|\, \textit{today is rain}) = \frac{2.5}{1 + 2.5} = 0.71$$

Rule 2 is also fired. The prior probability of *tomorrow is dry* is converted into the prior odds, but the evidence *today is rain* reduces the odds by a factor of 0.4. This, in turn, diminishes the probability of *tomorrow is dry* from 0.5 to 0.29:

$$O(\textit{tomorrow is dry}) = \frac{0.5}{1 - 0.5} = 1.0$$

$$O(\textit{tomorrow is dry} \,|\, \textit{today is rain}) = 0.4 \times 1.0 = 0.4$$

$$p(\textit{tomorrow is dry} \,|\, \textit{today is rain}) = \frac{0.4}{1 + 0.4} = 0.29$$

Hence if it is raining today there is a 71 per cent chance of it raining and a 29 per cent chance of it being dry tomorrow.

Further suppose that the user input is *today is dry*. By a similar calculation there is a 62 per cent chance of it being dry and a 38 per cent chance of it raining tomorrow.

Now we have examined the basic principles of Bayesian rules of evidence, we can incorporate some new knowledge in our expert system. To do this, we need to determine conditions when the weather actually did change. Analysis of the data provided in Table 3.3 allows us to develop the following knowledge base (the Leonardo expert system shell is used here).

Knowledge base

```
/* FORECAST: BAYESIAN ACCUMULATION OF EVIDENCE

control bayes

Rule: 1
if      today is rain {LS 2.5 LN .6}
then    tomorrow is rain {prior .5}

Rule: 2
if      today is dry {LS 1.6 LN .4}
then    tomorrow is dry {prior .5}

Rule: 3
if      today is rain
and     rainfall is low {LS 10 LN 1}
then    tomorrow is dry {prior .5}

Rule: 4
if      today is rain
and     rainfall is low
and     temperature is cold {LS 1.5 LN 1}
then    tomorrow is dry {prior .5}

Rule: 5
if      today is dry
and     temperature is warm {LS 2 LN .9}
then    tomorrow is rain {prior .5}

Rule: 6
if      today is dry
and     temperature is warm
and     sky is overcast {LS 5 LN 1}
then    tomorrow is rain {prior .5}

/* The SEEK directive sets up the goal of the rule set

seek tomorrow
```

Dialogue

Based on the information provided by the user, the expert system determines whether we can expect a dry day tomorrow. The user's answers are indicated by arrows. We assume that rainfall is low if it is less than 4.1 mm, the temperature is cold if the average daily temperature is lower than or equal to 7.0°C, and warm if it is higher than 7.0°C. Finally, sunshine less than 4.6 hours a day stands for overcast.

What is the weather today?
⇒ **rain**

Rule: 1
if today is rain {LS 2.5 LN .6}
then tomorrow is rain {prior .5}

$$O(\textit{tomorrow is rain}) = \frac{0.5}{1-0.5} = 1.0$$

$$O(\textit{tomorrow is rain} \mid \textit{today is rain}) = 2.5 \times 1.0 = 2.5$$

$$p(\textit{tomorrow is rain} \mid \textit{today is rain}) = \frac{2.5}{1+2.5} = 0.71$$

tomorrow is rain {0.71}

Rule: 2
if today is dry {LS 1.6 LN .4}
then tomorrow is dry {prior .5}

$$O(\textit{tomorrow is dry}) = \frac{0.5}{1-0.5} = 1.0$$

$$O(\textit{tomorrow is dry} \mid \textit{today is rain}) = 0.4 \times 1.0 = 0.4$$

$$p(\textit{tomorrow is dry} \mid \textit{today is rain}) = \frac{0.4}{1+0.4} = 0.29$$

tomorrow is rain {0.71}
dry {0.29}

What is the rainfall today?
⇒ **low**

Rule: 3
if today is rain
and rainfall is low {LS 10 LN 1}
then tomorrow is dry {prior .5}

$$O(\textit{tomorrow is dry}) = \frac{0.29}{1-0.29} = 0.41$$

$$O(\textit{tomorrow is dry} \mid \textit{today is rain} \cap \textit{rainfall is low}) = 10 \times 0.41 = 4.1$$

$$p(\textit{tomorrow is dry} \mid \textit{today is rain} \cap \textit{rainfall is low}) = \frac{4.1}{1+4.1} = 0.80$$

tomorrow is dry {0.80}
rain {0.71}

What is the temperature today?
⇒ **cold**

Rule: 4
if today is rain
and rainfall is low
and temperature is cold {LS 1.5 LN 1}
then tomorrow is dry {prior .5}

$$O(\textit{tomorrow is dry}) = \frac{0.80}{1-0.80} = 4$$

$$O(\textit{tomorrow is dry} \mid \textit{today is rain} \cap \textit{rainfall is low} \cap \textit{temperature is cold}) = 1.50 \times 4 = 6$$

$$p(\textit{tomorrow is dry} \mid \textit{today is rain} \cap \textit{rainfall is low} \cap \textit{temperature is cold}) = \frac{6}{1+6} = 0.86$$

tomorrow is dry {0.86}
rain {0.71}

Rule: 5
if today is dry
and temperature is warm {LS 2 LN .9}
then tomorrow is rain {prior .5}

$$O(\textit{tomorrow is rain}) = \frac{0.71}{1-0.71} = 2.45$$

$$O(\textit{tomorrow is rain} \mid \textit{today is not dry} \cap \textit{temperature is not warm}) = 0.9 \times 2.45 = 2.21$$

$$p(\textit{tomorrow is rain} \mid \textit{today is not dry} \cap \textit{temperature is not warm}) = \frac{2.21}{1+2.21} = 0.69$$

tomorrow is dry {0.86}
rain {0.69}

What is the cloud cover today?
⇒ **overcast**

Rule: 6
if today is dry
and temperature is warm
and sky is overcast {LS 5 LN 1}
then tomorrow is rain {prior .5}

$$O(\textit{tomorrow is rain}) = \frac{0.69}{1 - 0.69} = 2.23$$

$$O(\textit{tomorrow is rain} \mid \textit{today is not dry} \cap \textit{temperature is not warm} \cap \textit{sky is overcast}) = 1.0 \times 2.23 = 2.23$$

$$p(\textit{tomorrow is rain} \mid \textit{today is not dry} \cap \textit{temperature is not warm} \cap \textit{sky is overcast}) = \frac{2.23}{1 + 2.23} = 0.69$$

tomorrow is dry {0.86}
rain {0.69}

This means that we have two potentially true hypotheses, *tomorrow is dry* and *tomorrow is rain*, but the likelihood of the first one is higher.

From Table 3.3 you can see that our expert system made only four mistakes. This is an 86 per cent success rate, which compares well with the results provided in Naylor (1987) for the same case of the London weather.

3.5 Bias of the Bayesian method

The framework for Bayesian reasoning requires probability values as primary inputs. The assessment of these values usually involves human judgement. However, psychological research shows that humans either cannot elicit probability values consistent with the Bayesian rules or do it badly (Burns and Pearl, 1981; Tversky and Kahneman, 1982). This suggests that the conditional probabilities may be inconsistent with the prior probabilities given by the expert. Consider, for example, a car that does not start and makes odd noises when you press the starter. The conditional probability of the starter being faulty if the car makes odd noises may be expressed as

IF the symptom is 'odd noises'
THEN the starter is bad {with probability 0.7}

Apparently the conditional probability that the starter is not bad if the car makes odd noises is

$$p(\text{starter is not bad} \mid \text{odd noises}) = p(\text{starter is good} \mid \text{odd noises}) = 1 - 0.7 = 0.3$$

Therefore, we can obtain a companion rule that states

IF the symptom is 'odd noises'
THEN the starter is good {with probability 0.3}

Domain experts do not deal easily with conditional probabilities and quite often deny the very existence of the **hidden implicit probability** (0.3 in our example).

In our case, we would use available statistical information and empirical studies to derive the following two rules:

IF the starter is bad
THEN the symptom is 'odd noises' {with probability 0.85}

IF the starter is bad
THEN the symptom is not 'odd noises' {with probability 0.15}

To use the Bayesian rule, we still need the prior probability, the probability that the starter is bad if the car does not start. Here we need an expert judgement. Suppose the expert supplies us the value of 5 per cent. Now we can apply the Bayesian rule (3.18) to obtain

$$p(\text{starter is bad} \,|\, \text{odd noises}) = \frac{0.85 \times 0.05}{0.85 \times 0.05 + 0.15 \times 0.95} = 0.23$$

The number obtained is significantly lower than the expert's estimate of 0.7 given at the beginning of this section.

Why this inconsistency? Did the expert make a mistake?

The most obvious reason for the inconsistency is that the expert made different assumptions when assessing the conditional and prior probabilities. We may attempt to investigate it by working backwards from the posterior probability $p(\text{starter is bad} \,|\, \text{odd noises})$ to the prior probability $p(\text{starter is bad})$. In our case, we can assume that

$$p(\text{starter is good}) = 1 - p(\text{starter is bad})$$

From Eq. (3.18) we obtain

$$p(H) = \frac{p(H|E) \times p(E|\neg H)}{p(H|E) \times p(E|\neg H) + p(E|H)[1 - p(H|E)]}$$

where:

$p(H) = p(\text{starter is bad})$;

$p(H|E) = p(\text{starter is bad} \,|\, \text{odd noises})$;

$p(E|H) = p(\text{odd noises} \,|\, \text{starter is bad})$;

$p(E|\neg H) = p(\text{odd noises} \,|\, \text{starter is good})$.

If we now take the value of 0.7, $p(\text{starter is bad}|\text{odd noises})$, provided by the expert as the correct one, the prior probability $p(\text{starter is bad})$ would have

to be

$$p(H) = \frac{0.7 \times 0.15}{0.7 \times 0.15 + 0.85 \times (1 - 0.7)} = 0.29$$

This value is almost six times larger than the figure of 5 per cent provided by the expert. Thus the expert indeed uses quite different estimates of the prior and conditional probabilities.

In fact, the prior probabilities also provided by the expert are likely to be inconsistent with the likelihood of sufficiency, *LS*, and the likelihood of necessity, *LN*. Several methods are proposed to handle this problem (Duda *et al.*, 1976). The most popular technique, first applied in PROSPECTOR, is the use of a piecewise linear interpolation model (Duda *et al.*, 1979).

However, to use the subjective Bayesian approach, we must satisfy many assumptions, including the conditional independence of evidence under both a hypothesis and its negation. As these assumptions are rarely satisfied in real-world problems, only a few systems have been built based on Bayesian reasoning. The best known one is PROSPECTOR, an expert system for mineral exploration (Duda *et al.*, 1979).

3.6 Certainty factors theory and evidential reasoning

Certainty factors theory is a popular alternative to Bayesian reasoning. The basic principles of this theory were first introduced in MYCIN, an expert system for the diagnosis and therapy of blood infections and meningitis (Shortliffe and Buchanan, 1975). The developers of MYCIN found that medical experts expressed the strength of their belief in terms that were neither logical nor mathematically consistent. In addition, reliable statistical data about the problem domain was not available. Therefore, the MYCIN team was unable to use a classical probability approach. Instead they decided to introduce a **certainty factor** (***cf***), a number to measure the expert's belief. The maximum value of the certainty factor was +1.0 (definitely true) and the minimum −1.0 (definitely false). A positive value represented a degree of belief and a negative a degree of disbelief. For example, if the expert stated that some evidence was almost certainly true, a *cf* value of 0.8 would be assigned to this evidence. Table 3.4 shows some uncertain terms interpreted in MYCIN (Durkin, 1994).

In expert systems with certainty factors, the knowledge base consists of a set of rules that have the following syntax:

IF <evidence>
THEN <hypothesis> {*cf*}

where *cf* represents belief in hypothesis *H* given that evidence *E* has occurred.

The certainty factors theory is based on two functions: measure of belief $MB(H,E)$ and measure of disbelief $MD(H,E)$ (Shortliffe and Buchanan, 1975).

Table 3.4 Uncertain terms and their interpretation

Term	Certainty factor
Definitely not	−1.0
Almost certainly not	−0.8
Probably not	−0.6
Maybe not	−0.4
Unknown	−0.2 to +0.2
Maybe	+0.4
Probably	+0.6
Almost certainly	+0.8
Definitely	+1.0

These functions indicate, respectively, the degree to which belief in hypothesis H would be increased if evidence E were observed, and the degree to which disbelief in hypothesis H would be increased by observing the same evidence E.

The measure of belief and disbelief can be defined in terms of prior and conditional probabilities as follows (Ng and Abramson, 1990):

$$MB(H,E) = \begin{cases} 1 & \text{if } p(H) = 1 \\ \dfrac{max\,[p(H|E), p(H)] - p(H)}{max\,[1,0] - p(H)} & \text{otherwise} \end{cases} \tag{3.29}$$

$$MD(H,E) = \begin{cases} 1 & \text{if } p(H) = 0 \\ \dfrac{min\,[p(H|E), p(H)] - p(H)}{min\,[1,0] - p(H)} & \text{otherwise} \end{cases} \tag{3.30}$$

where:

$p(H)$ is the prior probability of hypothesis H being true;

$p(H|E)$ is the probability that hypothesis H is true given evidence E.

The values of $MB(H,E)$ and $MD(H,E)$ range between 0 and 1. The strength of belief or disbelief in hypothesis H depends on the kind of evidence E observed. Some facts may increase the strength of belief, but some increase the strength of disbelief.

How can we determine the total strength of belief or disbelief in a hypothesis?

To combine them into one number, the certainty factor, the following equation is used:

$$cf = \frac{MB(H,E) - MD(H,E)}{1 - min\,[MB(H,E), MD(H,E)]} \tag{3.31}$$

Thus *cf*, which can range in MYCIN from −1 to +1, indicates the total belief in hypothesis *H*.

The MYCIN approach can be illustrated through an example. Consider a simple rule:

IF *A* is *X*
THEN *B* is *Y*

Quite often, an expert may not be absolutely certain that this rule holds. Also suppose it has been observed that in some cases, even when the IF part of the rule is satisfied and object *A* takes on value *X*, object *B* can acquire some different value *Z*. In other words, we have here the uncertainty of a quasi-statistical kind.

The expert usually can associate a certainty factor with each possible value *B* given that *A* has value *X*. Thus our rule might look as follows:

IF *A* is *X*
THEN *B* is *Y* {*cf* 0.7};
B is *Z* {*cf* 0.2}

What does it mean? Where is the other 10 per cent?

It means that, given *A* has received value *X*, *B* will be *Y* 70 per cent and *Z* 20 per cent of the time. The other 10 per cent of the time it could be anything. In such a way the expert might reserve a possibility that object *B* can take not only two known values, *Y* and *Z*, but also some other value that has not yet been observed. Note that we assign multiple values to object *B*.

The certainty factor assigned by a rule is then propagated through the reasoning chain. Propagation of the certainty factor involves establishing the net certainty of the rule consequent when the evidence in the rule antecedent is uncertain. The net certainty for a single antecedent rule, $cf(H,E)$, can be easily calculated by multiplying the certainty factor of the antecedent, $cf(E)$, with the rule certainty factor, *cf*

$$cf(H,E) = cf(E) \times cf \tag{3.32}$$

For example,

IF the sky is clear
THEN the forecast is sunny {*cf* 0.8}

and the current certainty factor of *sky is clear* is 0.5, then

$$cf(H,E) = 0.5 \times 0.8 = 0.4$$

This result, according to Table 3.4, would read as 'It may be sunny.'

How does an expert system establish the certainty factor for rules with multiple antecedents?

For conjunctive rules such as

IF <evidence E_1>
AND <evidence E_2>
.
.
.
AND <evidence E_n>
THEN <hypothesis H> {cf}

the net certainty of the consequent, or in other words the certainty of hypothesis H, is established as follows:

$$cf(H, E_1 \cap E_2 \cap \ldots \cap E_n) = min\,[cf(E_1), cf(E_2), \ldots, cf(E_n)] \times cf \tag{3.33}$$

For example,

IF sky is clear
AND the forecast is sunny
THEN the action is 'wear sunglasses' {cf 0.8}

and the certainty of *sky is clear* is 0.9 and the certainty of the *forecast is sunny* is 0.7, then

$$cf(H, E_1 \cap E_2) = min\,[0.9, 0.7] \times 0.8 = 0.7 \times 0.8 = 0.56$$

According to Table 3.4, this conclusion might be interpreted as 'Probably it would be a good idea to wear sunglasses today.'

For disjunctive rules such as

IF <evidence E_1>
OR <evidence E_2>
.
.
.
OR <evidence E_n>
THEN <hypothesis H> {cf}

the certainty of hypothesis H is determined as follows:

$$cf(H, E_1 \cup E_2 \cup \ldots \cup E_n) = max\,[cf(E_1), cf(E_2), \ldots, cf(E_n)] \times cf \tag{3.34}$$

For example,

IF sky is overcast
OR the forecast is rain
THEN the action is 'take an umbrella' {*cf* 0.9}

and the certainty of *sky is overcast* is 0.6 and the certainty of the *forecast is rain* is 0.8, then

$$cf(H, E_1 \cup E_2) = max\,[0.6, 0.8] \times 0.9 = 0.8 \times 0.9 = 0.72$$

which can be interpreted as 'Almost certainly an umbrella should be taken today.'

Sometimes two or even more rules can affect the same hypothesis. How does an expert system cope with such situations?

When the same consequent is obtained as a result of the execution of two or more rules, the individual certainty factors of these rules must be merged to give a combined certainty factor for a hypothesis. Suppose the knowledge base consists of the following rules:

Rule 1: IF *A* is *X*
THEN *C* is *Z* {*cf* 0.8}

Rule 2: IF *B* is *Y*
THEN *C* is *Z* {*cf* 0.6}

What certainty should be assigned to object *C* having value *Z* if both Rule 1 and Rule 2 are fired? Our common sense suggests that, if we have two pieces of evidence (*A* is *X* and *B* is *Y*) from different sources (Rule 1 and Rule 2) supporting the same hypothesis (*C* is *Z*), then the confidence in this hypothesis should increase and become stronger than if only one piece of evidence had been obtained.

To calculate a combined certainty factor we can use the following equation (Durkin, 1994):

$$cf(cf_1, cf_2) = \begin{cases} cf_1 + cf_2 \times (1 - cf_1) & \text{if } cf_1 > 0 \text{ and } cf_2 > 0 \\ \dfrac{cf_1 + cf_2}{1 - min\,[|cf_1|, |cf_2|]} & \text{if } cf_1 < 0 \text{ or } cf_2 < 0 \\ cf_1 + cf_2 \times (1 + cf_1) & \text{if } cf_1 < 0 \text{ and } cf_2 < 0 \end{cases} \tag{3.35}$$

where:

cf_1 is the confidence in hypothesis H established by Rule 1;

cf_2 is the confidence in hypothesis H established by Rule 2;

$|cf_1|$ and $|cf_2|$ are absolute magnitudes of cf_1 and cf_2, respectively.

Thus, if we assume that

$$cf(E_1) = cf(E_2) = 1.0$$

then from Eq. (3.32) we get

$$cf_1(H, E_1) = cf(E_1) \times cf_1 = 1.0 \times 0.8 = 0.8$$
$$cf_2(H, E_2) = cf(E_2) \times cf_2 = 1.0 \times 0.6 = 0.6$$

and from Eq. (3.35) we obtain

$$\begin{aligned} cf(cf_1, cf_2) &= cf_1(H, E_1) + cf_2(H, E_2) \times [1 - cf_1(H, E_1)] \\ &= 0.8 + 0.6 \times (1 - 0.8) = 0.92 \end{aligned}$$

This example shows an incremental increase of belief in a hypothesis and also confirms our expectations.

Consider now a case when rule certainty factors have the opposite signs. Suppose that

$$cf(E_1) = 1 \text{ and } cf(E_2) = -1.0$$

then

$$cf_1(H, E_1) = 1.0 \times 0.8 = 0.8$$
$$cf_2(H, E_2) = -1.0 \times 0.6 = -0.6$$

and from Eq. (3.35) we obtain

$$cf(cf_1, cf_2) = \frac{cf_1(H, E_1) + cf_2(H, E_2)}{1 - min\,[|cf_1(H, E_1)|, |cf_2(H, E_2)|]} = \frac{0.8 - 0.6}{1 - min\,[0.8, 0.6]} = 0.5$$

This example shows how a combined certainty factor, or in other words net belief, is obtained when one rule, Rule 1, confirms a hypothesis but another, Rule 2, discounts it.

Let us consider now the case when rule certainty factors have negative signs. Suppose that

$$cf(E_1) = cf(E_2) = -1.0$$

then

$$cf_1(H, E_1) = -1.0 \times 0.8 = -0.8$$
$$cf_2(H, E_2) = -1.0 \times 0.6 = -0.6$$

and from Eq. (3.35) we obtain

$$\begin{aligned} cf(cf_1, cf_2) &= cf_1(H, E_1) + cf_2(H, E_2) \times [1 + cf_1(H, E_1)] \\ &= -0.8 - 0.6 \times (1 - 0.8) = -0.92 \end{aligned}$$

This example represents an incremental increase of disbelief in a hypothesis.

The certainty factors theory provides a **practical** alternative to Bayesian reasoning. The heuristic manner of combining certainty factors is different from the manner in which they would be combined if they were probabilities. The certainty theory is not 'mathematically pure' but does mimic the thinking process of a human expert.

To illustrate the evidential reasoning and the method of certainty factors propagation through a set of rules, consider again the FORECAST expert system developed in section 3.4.

3.7 FORECAST: an application of certainty factors

The expert system is required to predict whether it will rain tomorrow, or in other words to establish certainty factors for the multivalued object *tomorrow*. To simplify our task, we use the same set of rules as in section 3.4.

Knowledge base

```
/* FORECAST: AN APPLICATION OF CERTAINTY FACTORS

control cf

control 'threshold 0.01'

Rule: 1
if      today is rain
then    tomorrow is rain {cf 0.5}

Rule: 2
if      today is dry
then    tomorrow is dry {cf 0.5}

Rule: 3
if      today is rain
and     rainfall is low
then    tomorrow is dry {cf 0.6}

Rule: 4
if      today is rain
and     rainfall is low
and     temperature is cold
then    tomorrow is dry {cf 0.7}
```

Rule: 5
if today is dry
and temperature is warm
then tomorrow is rain {cf 0.65}

Rule: 6
if today is dry
and temperature is warm
and sky is overcast
then tomorrow is rain {cf 0.55}

seek tomorrow

Dialogue

To apply an inexact reasoning technique based on certainty factors, the expert system prompts the user to input not only the object value but also the certainty associated with this value. For example, using the Leonardo scale from 0 to 1.0, the following dialogue might be obtained:

What is the weather today?
⇒ **rain**

Rule: 1
if today is rain
then tomorrow is rain {cf 0.5}

$cf(\text{tomorrow is rain, today is rain}) = cf(\text{today is rain}) \times cf = 1.0 \times 0.5 = 0.5$

tomorrow is rain {0.50}

What is the rainfall today?
⇒ **low**

To what degree do you believe the rainfall is low? Enter a numeric certainty between 0 and 1.0 inclusive.
⇒ **0.8**

Rule: 3
if today is rain
and rainfall is low
then tomorrow is dry {cf 0.6}

$cf(\text{tomorrow is dry, today is rain} \cap \text{rainfall is low})$
$= min[cf(\text{today is rain}), cf(\text{rainfall is low})] \times cf = min[1, 0.8] \times 0.6 = 0.48$

tomorrow is rain {0.50}
dry {0.48}

What is the temperature today?
⇒ **cold**

To what degree do you believe the temperature is cold? Enter a numeric certainty between 0 and 1.0 inclusive.
⇒ **0.9**

Rule: 4
if today is rain
and rainfall is low
and temperature is cold
then tomorrow is dry {cf 0.7}

$$cf(\text{tomorrow is dry, today is rain} \cap \text{rainfall is low} \cap \text{temperature is cold}) = min[cf(\text{today is rain}), cf(\text{rainfall is low}), cf(\text{temperature is cold})] \times cf = min[1, 0.8, 0.9] \times 0.7 = 0.56$$

tomorrow is dry {0.56}
rain {0.50}

$$cf(cf_{\text{Rule:3}}, cf_{\text{Rule:4}}) = cf_{\text{Rule:3}} + cf_{\text{Rule:4}} \times (1 - cf_{\text{Rule:3}}) = 0.48 + 0.56 \times (1 - 0.48) = 0.77$$

tomorrow is dry {0.77}
rain {0.50}

Now we would conclude that the probability of having a dry day tomorrow is almost certain; however, we also may expect some rain!

3.8 Comparison of Bayesian reasoning and certainty factors

In the previous sections, we outlined the two most popular techniques for uncertainty management in expert systems. Now we will compare these techniques and determine the kinds of problems that can make effective use of either Bayesian reasoning or certainty factors.

Probability theory is the oldest and best-established technique to deal with inexact knowledge and random data. It works well in such areas as forecasting and planning, where statistical data is usually available and accurate probability statements can be made.

An expert system that applied the Bayesian technique, PROSPECTOR, was developed to aid exploration geologists in their search for ore deposits. It was very successful; for example, using geological, geophysical and geochemical data, PROSPECTOR predicted the existence of molybdenum near Mount Tolman in Washington State (Campbell *et al.*, 1982). But the PROSPECTOR team could rely on valid data about known mineral deposits and reliable statistical information. The probabilities of each event were defined as well. The PROSPECTOR

team also could assume the conditional independence of evidence, a constraint that must be satisfied in order to apply the Bayesian approach.

However, in many areas of possible applications of expert systems, reliable statistical information is not available or we cannot assume the conditional independence of evidence. As a result, many researchers have found the Bayesian method unsuitable for their work. For example, Shortliffe and Buchanan could not use a classical probability approach in MYCIN because the medical field often could not provide the required data (Shortliffe and Buchanan, 1975). This dissatisfaction motivated the development of the certainty factors theory.

Although the certainty factors approach lacks the mathematical correctness of the probability theory, it appears to outperform subjective Bayesian reasoning in such areas as diagnostics, particularly in medicine. In diagnostic expert systems like MYCIN, rules and certainty factors come from the expert's knowledge and his or her intuitive judgements. Certainty factors are used in cases where the probabilities are not known or are too difficult or expensive to obtain. The evidential reasoning mechanism can manage incrementally acquired evidence, the conjunction and disjunction of hypotheses, as well as evidences with different degrees of belief. Besides, the certainty factors approach provides better explanations of the control flow through a rule-based expert system.

The Bayesian approach and certainty factors are different from one another, but they share a common problem: finding an expert able to quantify personal, subjective and qualitative information. Humans are easily biased, and therefore the choice of an uncertainty management technique strongly depends on the existing domain expert.

The Bayesian method is likely to be the most appropriate if reliable statistical data exists, the knowledge engineer is able to lead, and the expert is available for serious decision-analytical conversations. In the absence of any of the specified conditions, the Bayesian approach might be too arbitrary and even biased to produce meaningful results. It should also be mentioned that the Bayesian belief propagation is of exponential complexity, and thus is impractical for large knowledge bases.

The certainty factors technique, despite the lack of a formal foundation, offers a simple approach for dealing with uncertainties in expert systems and delivers results acceptable in many applications.

3.9 Summary

In this chapter, we presented two uncertainty management techniques used in expert systems: Bayesian reasoning and certainty factors. We identified the main sources of uncertain knowledge and briefly reviewed probability theory. We considered the Bayesian method of accumulating evidence and developed a simple expert system based on the Bayesian approach. Then we examined the certainty factors theory (a popular alternative to Bayesian reasoning) and developed an expert system based on evidential reasoning. Finally, we compared

Bayesian reasoning and certainty factors, and determined appropriate areas for their applications.

The most important lessons learned in this chapter are:

- Uncertainty is the lack of exact knowledge that would allow us to reach a perfectly reliable conclusion. The main sources of uncertain knowledge in expert systems are: weak implications, imprecise language, missing data and combining the views of different experts.
- Probability theory provides an exact, mathematically correct, approach to uncertainty management in expert systems. The Bayesian rule permits us to determine the probability of a hypothesis given that some evidence has been observed.
- PROSPECTOR, an expert system for mineral exploration, was the first successful system to employ Bayesian rules of evidence to propagate uncertainties throughout the system.
- In the Bayesian approach, an expert is required to provide the prior probability of hypothesis H and values for the likelihood of sufficiency, LS, to measure belief in the hypothesis if evidence E is present, and the likelihood of necessity, LN, to measure disbelief in hypothesis H if the same evidence is missing. The Bayesian method uses rules of the following form:

  IF E is true {LS, LN}
  THEN H is true {*prior probability*}

- To employ the Bayesian approach, we must satisfy the conditional independence of evidence. We also should have reliable statistical data and define the prior probabilities for each hypothesis. As these requirements are rarely satisfied in real-world problems, only a few systems have been built based on Bayesian reasoning.
- Certainty factors theory is a popular alternative to Bayesian reasoning. The basic principles of this theory were introduced in MYCIN, a diagnostic medical expert system.
- Certainty factors theory provides a judgemental approach to uncertainty management in expert systems. An expert is required to provide a certainty factor, cf, to represent the level of belief in hypothesis H given that evidence E has been observed. The certainty factors method uses rules of the following form:

  IF E is true
  THEN H is true {cf}

- Certainty factors are used if the probabilities are not known or cannot be easily obtained. Certainty theory can manage incrementally acquired evidence, the conjunction and disjunction of hypotheses, as well as evidences with different degrees of belief.

- Both Bayesian reasoning and certainty theory share a common problem: finding an expert able to quantify subjective and qualitative information.

Questions for review

1 What is uncertainty? When can knowledge be inexact and data incomplete or inconsistent? Give an example of inexact knowledge.

2 What is probability? Describe mathematically the conditional probability of event *A* occurring given that event *B* has occurred. What is the Bayesian rule?

3 What is Bayesian reasoning? How does an expert system rank potentially true hypotheses? Give an example.

4 Why was the PROSPECTOR team able to apply the Bayesian approach as an uncertainty management technique? What requirements must be satisfied before Bayesian reasoning will be effective?

5 What are the likelihood of sufficiency and likelihood of necessity? How does an expert determine values for *LS* and *LN*?

6 What is a prior probability? Give an example of the rule representation in the expert system based on Bayesian reasoning.

7 How does a rule-based expert system propagate uncertainties using the Bayesian approach?

8 Why may conditional probabilities be inconsistent with the prior probabilities provided by the expert? Give an example of such an inconsistency.

9 Why is the certainty factors theory considered as a **practical** alternative to Bayesian reasoning? What are the measure of belief and the measure of disbelief? Define a certainty factor.

10 How does an expert system establish the net certainty for conjunctive and disjunctive rules? Give an example for each case.

11 How does an expert system combine certainty factors of two or more rules affecting the same hypothesis? Give an example.

12 Compare Bayesian reasoning and certainty factors. Which applications are most suitable for Bayesian reasoning and which for certainty factors? Why? What is a common problem in both methods?

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